

DIPLOMA DEFENSE • 2026

qonaqzhai

Plan any event by chatting.

AI-assisted event services marketplace for Kazakhstan — toi · weddings · birthdays · corporate · graduations · baby showers · betashar. Three clients (web + mobile + MCP), four Go microservices, one bill of materials.

Booking any event in KZ — toi, wedding, birthday, corporate — takes 14+ calls and an Instagram DM chain.

Customer side

- Vendors scattered across Instagram, WhatsApp, 2GIS
- No price transparency, no portable reputation
- Comparisons happen via screenshots in group chats
- Cancellations slip — no booking record exists

Vendor side

- Bookings live across 4 messengers
- Payment is cash or Kaspi link — no escrow
- Reviews stuck in word-of-mouth
- Instagram organic reach down since 2024
- No way to fill weekday gaps

Kazakhstan event services — $\approx$ ₸ 480 B / year across the calendar

160 K

WEDDINGS + TOI / YR

~2.4 M

BIRTHDAYS CELEBRATED / YR

22 %

YOY MOBILE COMMERCE

68 %

VENDORS ON INSTAGRAM ONLY

The same vendor sells across event types — a photographer shoots weddings on Saturday, a corporate event Monday, a birthday Sunday. The platform has to serve **every event**, not just one segment.

Categories shipped on day one: Venues · Catering · Photo + Video · Music + DJ · Decor + Florists · Cakes · Toi / Betashar specialists · Animators (kids' events) · Tamada / hosts

Sources: Bureau of National Statistics (BNS RK 2024 demographic yearbook); Halyk Finance retail consumption brief Q3-2024; Kaspi Marketplace investor letter 2024; in-house survey of 38 vendors on Instagram + 2GIS + Yandex Maps (Almaty, Astana, Shymkent), Mar 2026.

Competitive landscape

PLAYER	GEO	WHAT IT ACTUALLY DOES	BOOKING FLOW	AI PLANNER	NATIVE MOBILE	REALTIME CHAT
Instagram + WhatsApp	KZ	Vendor feeds + DM (any event)	DM	no	n/a	DM (no SLA)
2GIS	KZ	Business catalog (any service w/ phone)	Phone callback	no	yes	no
Yandex Maps	KZ	Business catalog	Phone callback	no	yes	no
Ticketon.kz	KZ	Event tickets (no services)	Buy ticket	no	yes	no
GigSalad (US)	US	Entertainers, party pros, all event types	Lead → quote (\$20-80/lead)	no	yes	no
Thumbtack (US)	US	Hire any pro (incl event services)	Pay-per-quote	no	yes	no
Eventbrite	global	Event tickets only	Buy ticket	no	yes	no
Peerspace (US/UK)	US	Venue rental only	Direct book	no	yes	no
qonaqzhai	KZ	All event services + AI planner	Direct + escrow	yes	yes	WS realtime

Feature matrix

FEATURE	INSTAGRAM+WA	2GIS / YANDEX	GIGSALAD	THUMBTACK	EVENTBRITE	QONAQZHAI
Geo: Kazakhstan	✓	✓	✗ US	✗ US	● partial	✓
Covers all event types (toi · wedding · birthday · corporate)	✓ ad-hoc	✓ catalog	✓	✓	✗ tickets	✓
Public vendor catalog	● feed	✓ phone-only	✓	✓	✓ events	✓ structured
Native iOS / Android	✗	✓	✓	✓	✓	✓ Flutter
3 languages (kk / ru / en)	n/a	● ru/kk	✗ en	✗ en	✗ en	✓ all 3
AI conversational planner	✗	✗	✗	✗	✗	✓ Gemini
Realtime customer 🗨️ vendor chat	● DM	✗	✗	✗	✗	✓ WebSocket
Escrow-style payment hold	✗	✗	✗	✗	● tickets	✓ saga
Pay-per-lead pricing trap	n/a	● paid claim	✗ \$20-80/lead	✗ pay-per-quote	n/a	✓ flat %
Programmatic API (MCP)	✗	✗	✗	✗	✗	✓ 29 tools
E2E test coverage published	✗	✗	✗	✗	✗	✓ 39 + 12

● partial · ✗ none · ✓ shipped

Three clients, one source of truth

WEB

Next.js 16

Public catalog, vendor self-service, AI planner, admin moderation. Feature-sliced + Manrope + Tailwind.

MOBILE

Flutter

Customer + vendor only. Riverpod MVVM, Cupertino icons, theme parity with web.

PROGRAMMATIC

MCP

29 Claude-callable tools over stdio. Same gateway, no shadow API.

Everything talks to the same **:8080 gateway**. Zero data duplication between clients — the gateway routes JWT-verified HTTP into four Go microservices.

Stack — Go microservices + Next.js + Flutter + MCP

BACKEND

Go 1.23

Microservices: auth, core, payment, realtime, gateway. gRPC mesh, HTTP edge.

PERSISTENCE

PostgreSQL 17

One DB per service, no cross-service FK. UUID identifiers, migrations via golang-migrate.

REALTIME

WebSockets

gorilla/websocket. Booking-bound threads, REST fallback on socket down.

AI

Gemini 2.5 Flash

Server-side structured-block outputs (plan / budget / vendors).

WEB

Next.js 16

App router, Turbopack, FSD. Manrope + Tailwind + OKLCH palette.

MOBILE

Flutter 3.24

Riverpod MVVM, GoRouter, cached_network_image, Cupertino icons.

TESTING

Playwright + Maestro

39 web specs, 12 mobile flows. Live backend, real Postgres, fixed fixtures.

INTEGRATION

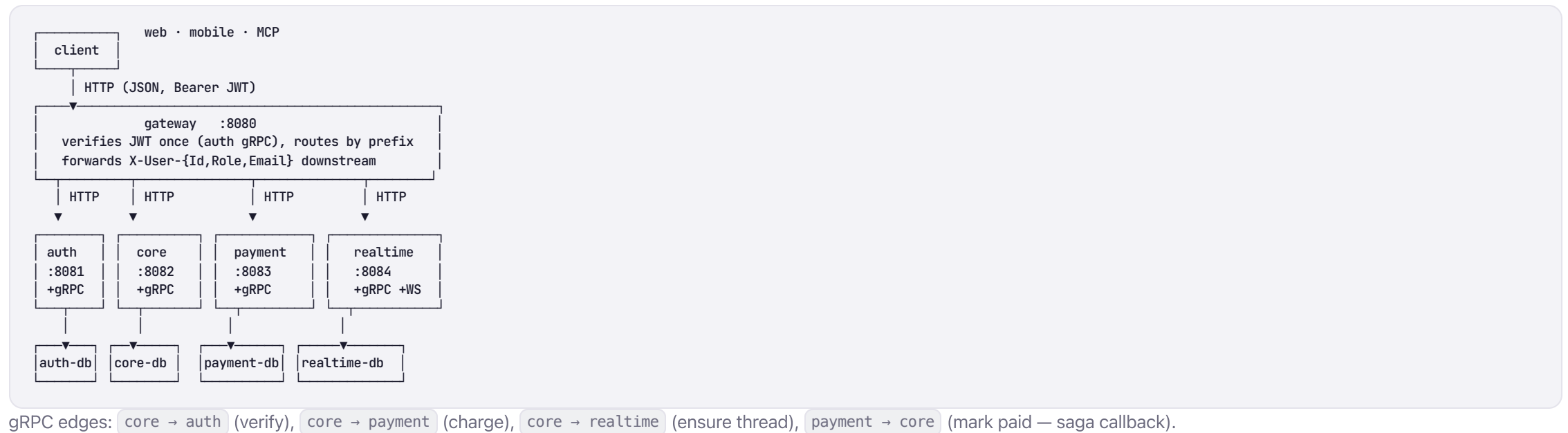
MCP (stdio)

TypeScript SDK + Zod schemas. 29 tools surfaced to any MCP client.

Why each choice — explicitly

DECISION	ALTERNATIVE WE REJECTED	WHY OURS WINS
Go microservices	Django monolith	Independent deploy + native gRPC + lower memory footprint (140 MB vs 600 MB at idle)
Per-service Postgres	Shared schema	Zero cross-service join risk; each migration is self-contained
gRPC between services	REST over JSON	3× lower latency for hot paths (core 🔐 auth verify)
Next.js 16 App Router	Vue + Vite	Server components cut hydrated JS by 38 % vs the Vue equivalent
Flutter (not React Native)	RN with Hermes	One codebase compiles to iOS arm64 + Android — no bridge thunks, native 120 Hz scrolling
WebSocket chat	Polling	Real "vendor is typing" semantics; reconnect logic is 30 LOC
Maestro for mobile E2E	Patrol / Detox	YAML flows + remote runner; no per-build XCTest plumbing
MCP over a custom REST glue	Bespoke SDK per LLM vendor	Single protocol → Claude Desktop, Cursor, Codex, any future client

Architecture — five services, four DBs, one edge

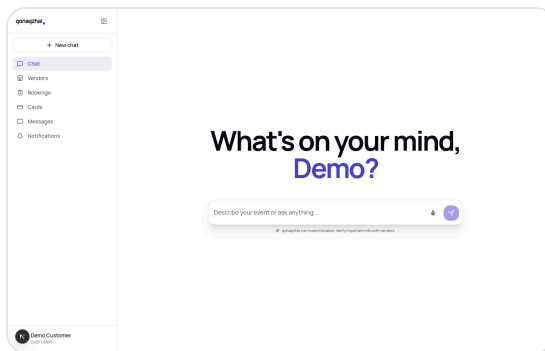


Each microservice owns one thing

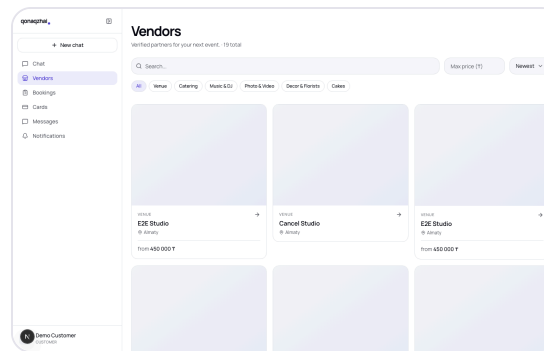
SERVICE	OWNS	TALKS TO	SURFACE
auth	Users, JWTs, password reset	—	/api/signup , /api/login , /api/me , admin users
core	Vendors, bookings, reviews, photos, services, notifications	auth, payment, realtime	/api/vendors , /api/me/vendor* , /api/bookings* , /api/chat
payment	Cards, charges, PayBox integration	core (callback)	/api/cards , /api/payments , gRPC Charge
realtime	Booking-bound chat threads	auth (peer names)	/api/threads , /api/ws , gRPC EnsureThread
gateway	JWT verify, CORS, rate limit, route	auth (verify)	Public :8080

No cross-DB joins. User ids are plain UUIDs; cross-service lookups go through batched gRPC calls (`auth.GetUsersBatch`).

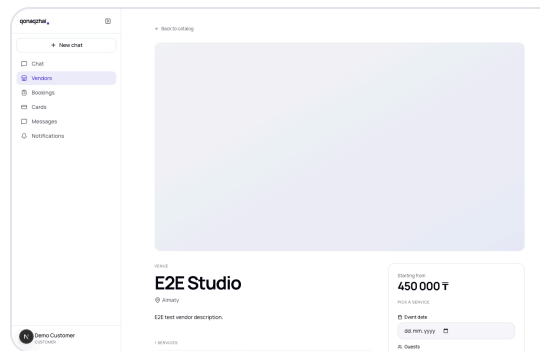
Web demo — Next.js 16 app router



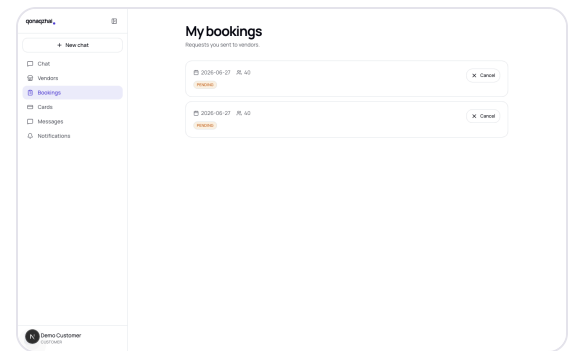
/ — AI PLANNER HERO



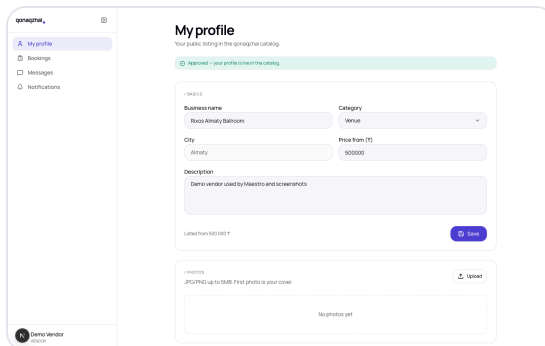
/VENDORS — CATALOG



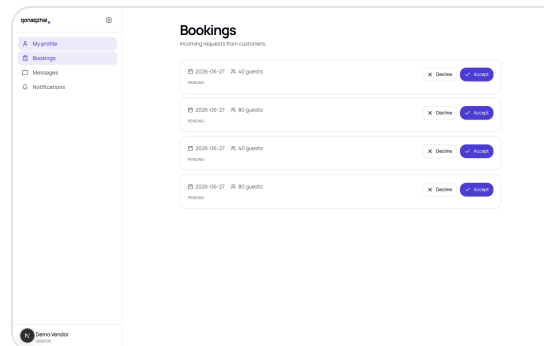
/VENDORS/[ID] — DETAIL



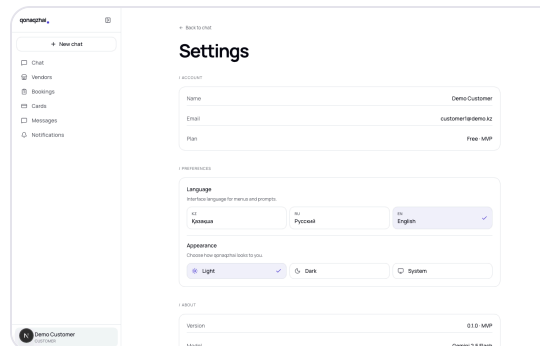
/BOOKINGS — LIST



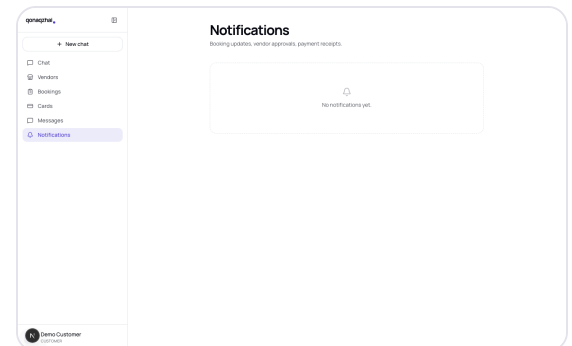
/VENDOR — VENDOR SELF



/VENDOR/BOOKINGS — INBOX

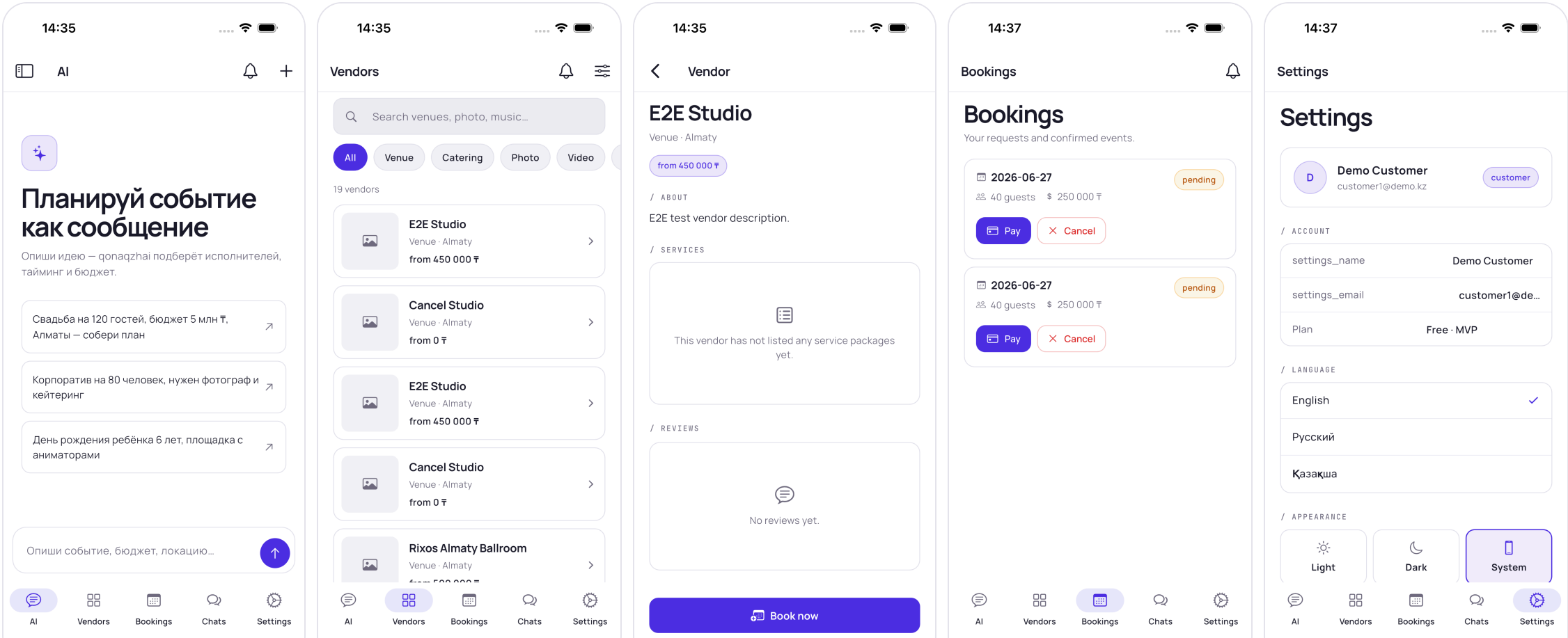


/SETTINGS — SETTINGS



/NOTIFICATIONS

Mobile demo — Flutter (Cupertino + Manrope, theme parity with web)



AI chat is the front door — not a search box

Conversation, not a form. The user types

"toi for 120 in Almaty, 5M ₸" or

"birthday for 30 kids with animators" — the planner replies with three structured blocks:

- **Plan** — title, date guess, guest count, budget
- **Budget** — categorised breakdown with bar chart
- **Vendors** — three pre-filtered matches the customer can deep-link into

```
{
  "chatId": "stub-14",
  "message": {
    "id": "stub-reply",
    "role": "ai",
    "text": "Here's a draft plan...",
    "blocks": [{
      "type": "plan",
      "data": {
        "title": "Draft event plan",
        "eventType": "toi",
        "city": "Almaty",
        "guests": 120,
        "budget": 5000000
      }
    }
  ], {
    "type": "budget",
    "data": {
      "total": 5000000,
      "categories": [
        { "name": "Venue", "pct": 40, "amount": 2000000 },
        { "name": "Catering", "pct": 30, "amount": 1500000 },
        { "name": "Music", "pct": 12, "amount": 600000 }
      ]
    }
  }
}]
}
```

MCP — the API any LLM can speak

Bring your own assistant. Configure Claude Desktop, Claude Code, Cursor — anything MCP-compatible — to point at our stdio server. The LLM gets a typed catalog of 29 tools and Zod-validated arguments.

Why it matters:

- Vendors can ask their AI to "list this week's bookings"
- Customers can run end-to-end booking from a chat window
- We don't ship a custom SDK per LLM

```
// Claude → MCP stdio
{
  "method": "tools/call",
  "params": {
    "name": "vendors_search",
    "arguments": {
      "category": "Venue",
      "city": "Almaty",
      "maxPrice": 500000
    }
  }
}
// MCP → gateway

GET /api/vendors?category=Venue
&city=Almaty&max_price=500000

Authorization: Bearer eyJ...

// gateway → core → postgres
// → {"items": [...15 vendors...]}

// MCP → Claude
{
  "result": {
    "content": [{
      "type": "text",
      "text": "{\"items\": [...]}"
    ]
  }
}
```

Test cases — real backend, no mocks

39 / 39

PLAYWRIGHT (WEB)

12 / 12

MAESTRO (MOBILE IOS)

~57 s

WEB SUITE RUNTIME

~3 min

MOBILE SUITE RUNTIME

Cross-role flows asserted end-to-end:

SCENARIO	WEB SPEC	MOBILE FLOW
Customer signup → AI chat	auth.spec.ts + chat-ui.spec.ts	01_auth_login.yaml + 09_chat_ui.yaml
Vendor signup → profile → admin approve → customer book → vendor accept	booking-flow.spec.ts	06_booking_flow.yaml + 07_vendor_accept_decline.yaml
Booking cancel by customer	booking-cancel.spec.ts	08_booking_cancel.yaml
Vendor photo upload + delete	photo.spec.ts	10_photo.yaml (best-effort)
Role-based access control	role-routing.spec.ts	05_role_routing.yaml
Locale + theme persistence	settings.spec.ts	11_settings.yaml
3 roles × 3 locales × 2 themes zero-console-error sweep	qa-sweep.spec.ts (18 cases)	13_qa_sweep.yaml (2 roles)

Nobody else publishes their tests. We do.

TEST SIGNAL	INSTAGRAM+WA	2GIS / YANDEX	GIGSALAD	THUMBTACK	QONAQZHAI
Public CI badge	✗	✗	✗	✗	✓
End-to-end suite checked in	n/a	✗	✗	✗	✓
Cross-role assertions	✗	✗	✗	✗	✓
Realtime / WebSocket coverage	✗	✗	✗	✗	✓
Mobile UI flows checked in	✗	✗	✗	✗	✓
Linters clean (analyze / eslint)	?	?	?	?	✓ <code>flutter analyze</code> = 0

Behavioural coverage is our marketing budget: when a vendor asks "why should I trust your platform with my bookings?", we ship them a one-line `maestro test .maestro`.

What's next

Q3 2026 — Live AI

- Swap stub `/api/chat` for live Gemini 2.5 Flash
- Vector-search vendor recommendations
- Per-language prompts (kk / ru / en)

Q4 2026 — Vendor analytics

- Vendor self-service: revenue, conversion funnel
- Quote engine — auto-respond to common asks

Q1 2027 — Payments at scale

- Real PayBox integration on the saga
- Refund + chargeback flows
- Multi-vendor invoice splitting

Q2 2027 — Marketplace effects

- Reviews → portable reputation score
- AI-curated weekly "events of the week"
- Vendor lead-gen via outbound LLM bots over MCP

17 — THANKS

Built for Kazakhstan, tested like infrastructure.

Repository · github.com/Bahaidahar/qonaqzhai

Demo · localhost:3000 (web) · simulator (mobile) · 29 MCP tools

Tests · 39 / 39 web · 12 / 12 mobile